National University of Computer and Emerging Sciences

Technical & Business Writing (2007)

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Course Instructor(s)

Nazia Imam, Javeria Ali, Sharmeen Ismail

Final Exam

Total Time: 3 Hours
Total Marks: 100
Total Questions: 03

Semester: SP-2025
Campus: Karachi
Dept: Computer Science

_____	_____	_____	_____
Student Name	Roll No	Section	Student Signature

CLO #2: Compose reports for effective performance in professional life.

Question 1 A: You are part of a tech startup that has developed a solar-powered charging station for public parks. Your product is portable, eco-friendly, and can charge up to 10 devices at a time. You believe this innovation would benefit your city's public spaces by promoting green energy and offering convenience to citizens. **[30 marks]**

Write an unsolicited technical proposal to the City Parks and Recreation Department, introducing your product and recommending its installation in major public parks. Write only the content asked in the question. Your proposal should include the following sections:

1. Statement of problem
2. Background
3. Need
4. Benefits
5. Limitations
6. Scope
7. Proposed Plan

Question 1 B: Read the given scenarios (1 to 8) and write in your copy if the given statements are true or false: [5 marks]

1. A company submits a proposal to a government agency in response to a detailed request for a new community health application. This proposal is an example of an unsolicited proposal.
2. A department head writes a proposal to the finance team within the same company requesting funds for new equipment. This is an example of an internal proposal.
3. A start-up sends a proposal to a potential investor without being asked, but it fails to explain the need for funding or market analysis. Background information is unnecessary in unsolicited proposals.
4. A proposal ends by summarizing the advantages of the project and encouraging the reader to take action. This is an effective and expected way to conclude a proposal.
5. A proposal submitted to a corporate client does not include any cost or timeline details. Cost estimation and timeline are optional in external proposals.
6. A government employee prepares a proposal for restructuring the city's emergency response plan. The proposal outlines logistics, staffing, and expected outcomes. This is an example of a planning proposal.
7. A team of scientists applies for funding to explore a new vaccine through an international health organization. This situation describes a research grant proposal.

8. A marketing team writes a proposal outlining a promotional strategy to boost brand awareness, which they send to their own CEO for review and approval. This proposal qualifies as a sales proposal.
   Choose the correct option and write in your answer copy:
9. **In the introduction of a survey report, you should include:**
   A. The purpose and scope of the survey B. A summary of the results
   C. A list of recommendations D. Your personal opinion
10. **What should the conclusion of a survey report ideally include?**
   A. Only the graphs B. Personal experiences
   C. Key findings and possible implications D. Irrelevant information

Question 1C: Match the type of feasibility with the correct project scenario. [2 marks]

Column A (Feasibility Type)	Column B (Scenario)
a. Technical Feasibility	1. A new bug tracking system is proposed, but employees resist using it.
b. Economic Feasibility	2. The company wants to launch a student data tracking app but needs to ensure it meets privacy regulations.
c. Operational Feasibility	3. A university wants to build a system using block chain, but has no in-house block chain experts.
d. Legal feasibility	4. The organization needs to evaluate whether the projected profits justify the cost of a new e-commerce platform.

Question 1 D: The project: Smart Waste Management System Using IoT and Mobile App Integration is being developed by a group of final-year BS Computer Science students at FAST as a part of their course project for the Software Engineering course. The project supervisor, Dr. Amira Khan, has asked the team to submit a second progress report to monitor technical development, challenges, and adherence to the proposed timeline.
With increasing urban population, waste management has become a growing challenge. Overflowing bins, irregular garbage collection, and inefficient truck routing lead to health risks and environmental issues. In response, this student-team proposed a Smart Waste Management System to address these problems using IoT sensors**,** GPS tracking**,** and a mobile application. Following are the project objective:

1. Use IoT sensors to detect bin fill levels.
2. Build a mobile app for real-time monitoring by municipal staff.
3. Develop a dashboard for visualizing data and scheduling.
4. Integrate GPS tracking and implement route optimization**.**
5. Ensure scalability for deployment in larger city zones.

Write the second progress report that: [25 marks]

- Allows reporting on progress in hardware deployment, mobile app enhancements, and data handling.
- Offers room for reflection on technical issues, user feedback, and changes in scope.
- Gives a platform to describe next milestones, such as ML integration or performance evaluation.

Note: According to the appropriate audience, follow the correct format for this report.

CLO #3: **Design, document, and develop a research project.**

Question 2 A: Write an informative abstract of 200-250 words for the given report. [20 marks]
Note: Do not copy sentences from the report.

A class of 100 university students was surveyed on their preferred study environment for exam preparation. The options were: library, home, or group study sessions.

Introduction:

Effective exam preparation is a key concern for university students, and the environment in which they study can significantly impact their academic performance. However, students have diverse preferences when it comes to study environments, ranging from quiet individual settings to interactive group sessions. Without clear data, it is difficult for university administrators to allocate resources and support services that truly reflect students' needs.

The objective of this survey was to identify and analyze students' preferred study environments for exam preparation, understand the reasons behind their choices, and provide insights that can help improve academic support structures at the university.
To achieve this, a survey was conducted among a class of 100 university students**,** asking them to indicate their preferred study environment from the following options: studying at home, studying in the library, or participating in group study sessions. Students were also asked to provide brief reasons for their choices. The responses were collected, categorized, and visually presented through graphs to facilitate analysis.
Findings:
According to the survey results:

- 45% of students preferred studying at home.
- 35% chose the library as their ideal study environment.
- 20% favored group study sessions.

Discussion:

- Home was favored for its comfort, flexibility, and absence of time restrictions. Many students appreciated the quiet and personal space it offers.
- Those who preferred the library cited fewer distractions, access to academic resources, and a focused atmosphere as their main reasons.
- Students who chose group study highlighted peer interaction, collaborative problem-solving, and motivational support as key benefits, although some admitted it can sometimes lead to off-topic conversations.

The library, preferred by 42% of the students, was chosen primarily for its quiet and structured atmosphere. Many respondents mentioned that the absence of distractions and the availability of academic resources such as books, internet access, and study desks made the library an ideal environment for focused preparation.

Studying at home was the choice for 38% of the students. Those who selected this option cited comfort, flexibility in scheduling, and the ability to personalize their study space as major reasons. However, some acknowledged that home environments could also introduce distractions, such as family interruptions or access to entertainment.

Only 20% of students preferred group study sessions. Students who favored this method believed it helped them stay motivated and understand difficult concepts through peer discussion. However, a larger portion of students avoided group study due to concerns about time wastage, off-topic discussions, and the difficulty of coordinating with others.
These results suggest that while students value interaction and peer support, they are more likely to choose environments that promote individual focus and control over study conditions.
Conclusion and Recommendations:
The survey shows that while students value different environments based on their study habits, a significant number benefit from quiet, independent study either at home or in the library. Universities could consider extending library hours during exam periods and offering more quiet zones. At the same time, organizing structured group study sessions with facilitators may help support those who thrive in collaborative environments.

Question 2 B: Arrange the steps in a typical Computer Science research process in the correct order: [2 marks]

a. Data collection
b. Problem formulation
c. Interpretation of results
d. Literature review

Question 2 C: Arrange the steps a researcher might follow when conducting research on a new software development model: [2 marks]

a. Test the model in a controlled environment
b. Design the model

c. Collect user feedback
d. Analyze feedback and refine the model

CLO #4: Displays the internationally accepted standards of technical communication

Question 3 A: Choose the right option and write only the option number in your copy: [4 marks]

1. **What should you do before sending a formal email with an attachment?**
   A. send it without writing anything in the body
   B. assume the recipient will find it themselves
   C. compress it without telling them
   D. mention the attachment in the body and make sure it is attached
2. **When sending a report as part of your email, how should it be included?**
   A. Pasted in the email body randomly
   B. Upload to social media and linked
   C. Sent as a file attachment with a reference to it in the email body
   D. Attached without mentioning it
3. **Read the email snippet below:**

   To: Dr. Asim (Supervisor)
   CC: Group Members
   BCC: Department Admin
   Subject: Submission of Final Year Project Draft

   Dear Dr. Asim,
   Attached is our group's draft for the final year project report. Kindly review and let us know if any improvements are needed.

   Regards,
   Sara Iqbal
   BSCS-7A

   What is the purpose of using BCC in this email?

   A. to hide the main recipient
   B. to keep the admin informed without showing it to others
   C. avoid sending the email to group members

4. You are sending an email to your professor and want to formally include another faculty member for awareness. Which field should you use?

A. To B. BCC C. CC D. Subject

Question 3 B: The following email content contains 5 basic mistakes. Write an improved version of it: [10 marks]

Subject: SUBMISSION OF ASSIGNMENT

Hi Sir,

i hope u r doing well. The assignment was a bit tricky, but I have completed it and attached it here.

Ali